

1	GMSS/2025/I/11
	The diagram shows the graph of $y = -x^2 + 8x - 12$. The curve cuts the x-axis at the points A and B. (a) Find the coordinates of A and B. [2] (b) Find the coordinates of the maximum point. [1] (c) Explain why the equation $-x^2 + 8x - 12 = k$ has no solution for some k values. [1]
2	GMSS/2025/I/16
	The sketch shows the graph of $y = ka^x$, where $a > 0$. The points $P(0, 3)$ and $Q(-3, 192)$ lie on the graph. (a) Find the value of a and of k. (b) R has coordinates $(-1, 8)$. Rick claims that points P, Q and R are collinear. Is he correct? Give a reason for your answer. [2]

3

YTSS/2025/II/04

(a) Complete the table of values for $y = \frac{x^2}{3} + \frac{2}{x} - 1$.
Values are given to one decimal place where appropriate.

x	-3	-2	-1.2	-1	0.25	0.5	1	1.5	2	3	4
y		-0.7	-2.2	-2.7	7.0	3.1	1.3	1.1	1.3	2.7	4.8

[1]

(b) On the grid, draw the graph of $y = \frac{x^2}{3} + \frac{2}{x} - 1$ for $-3 \leq x \leq 4$.

[3]

(c) The equation $\frac{x^2}{3} + \frac{2}{x} - 1 = p$ has only one solution. State the range of values of p .

[1]

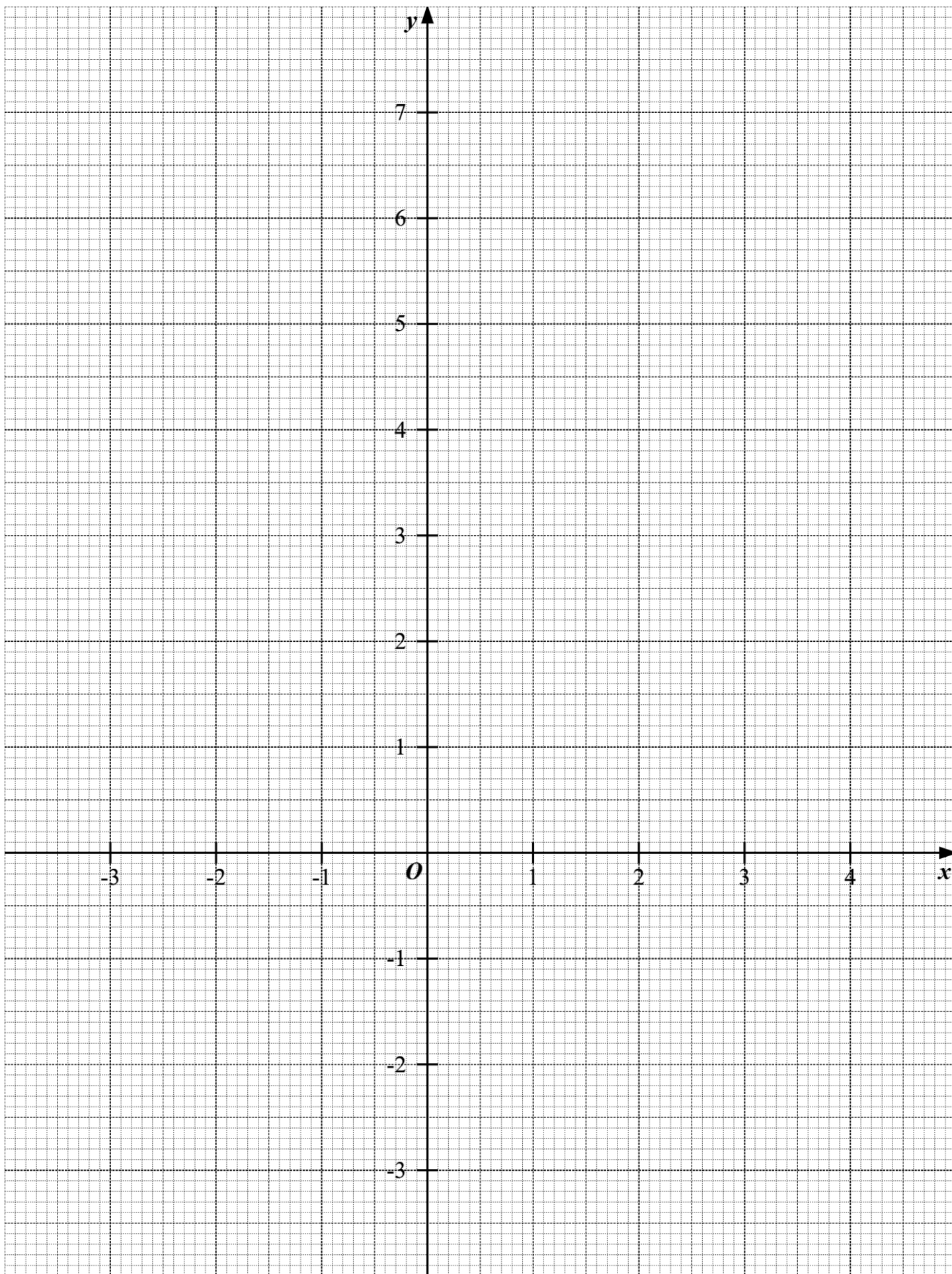
(d) The equation $2x^3 - 3x^2 - 12x + 12 = 0$ can be solved by inserting a straight line on the graph.

(i) State the equation of the straight line.

[2]

(ii) Hence, solve the equation $2x^3 - 3x^2 - 12x + 12 = 0$.

[4]



- (a) Complete the table of values for $y = -\frac{2x^3}{5} + 2x^2 - 3$.

x	-1.5	-1	0	1	1.5	2	3	3.5	4
y		-0.6	-3	-1.4	0.15	1.8	4.2	4.35	3.4

[1]

- (b) On the grid, draw the graph of $y = -\frac{2x^3}{5} + 2x^2 - 3$ for $-1.5 \leq x \leq 4$.

[3]

- (c) (i) On the same grid, draw the graph of $2y - 3x = 4$ for $-2 \leq x \leq 2$.

[2]

- (ii) Q is a point on the curve such that the gradient of the curve at Q is the same as the gradient of the line $2y - 3x = 4$.

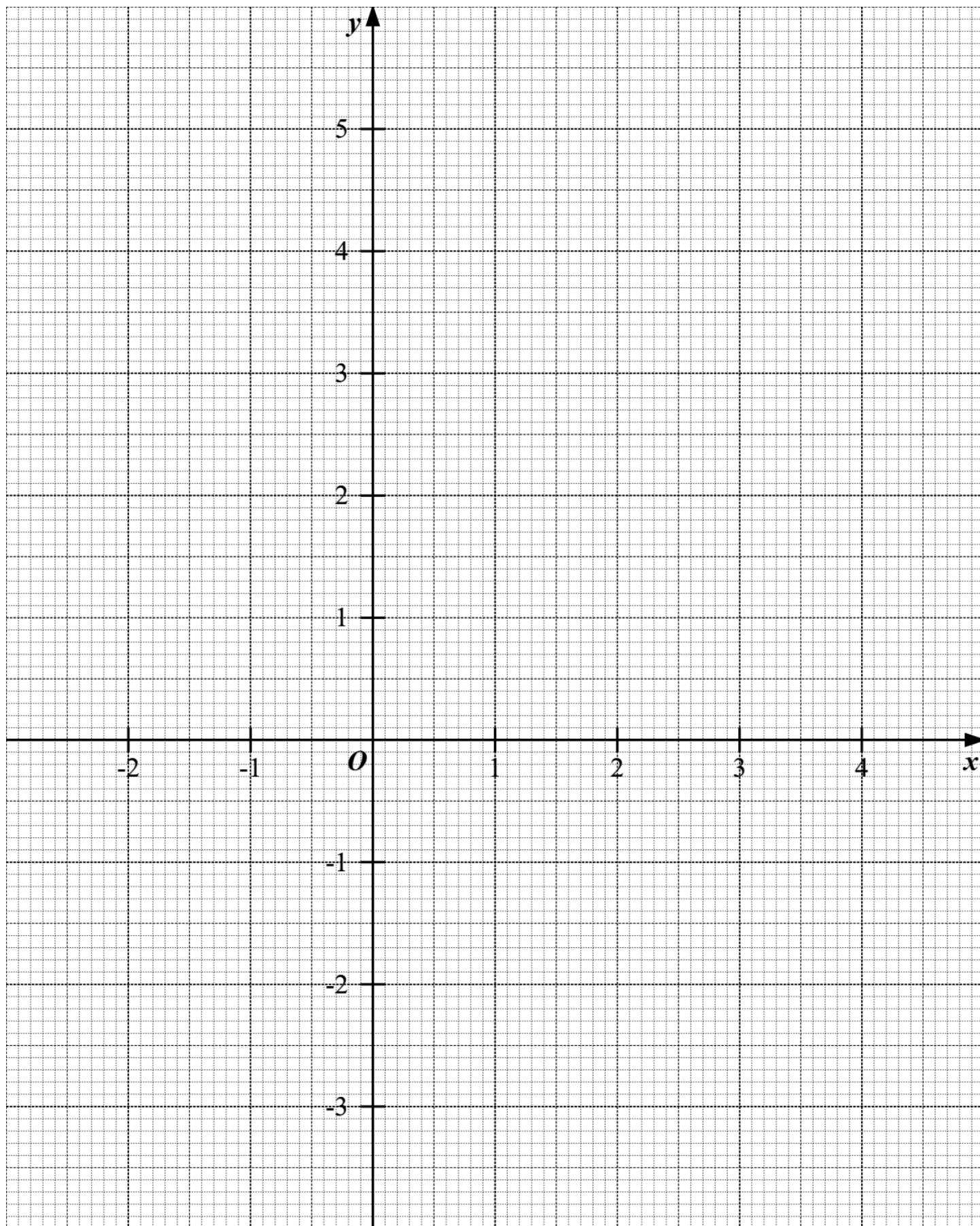
Using your graph, write down a possible x -coordinate of Q .

[1]

- (d) The equation $2x^3 - 10x^2 + 35 = 0$ can be solved by drawing a suitable straight line on the grid.

Find the equation of the straight line.

[2]



(a)

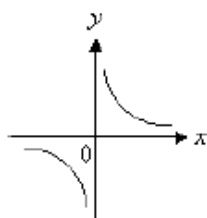


Figure 1

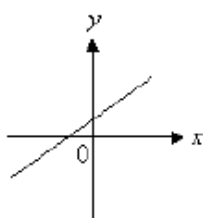


Figure 2

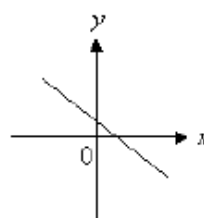


Figure 3

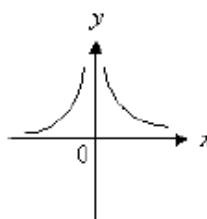


Figure 4

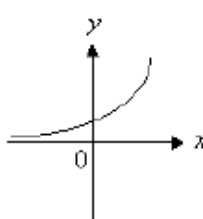


Figure 5

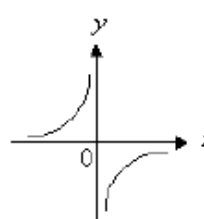


Figure 6

Which of the graphs shown above could be the graph of

(i) $y = 2^x$,

[1]

(ii) $y = \frac{3}{x^2}$,

[1]

(iii) $x + y = 1$?

[1]

(b) (i) Complete the table of values for $y = x\left(49 - \frac{x^2}{4}\right)$.

x	0	2	4	6	8	10	12
y	0	96	180		264	240	156

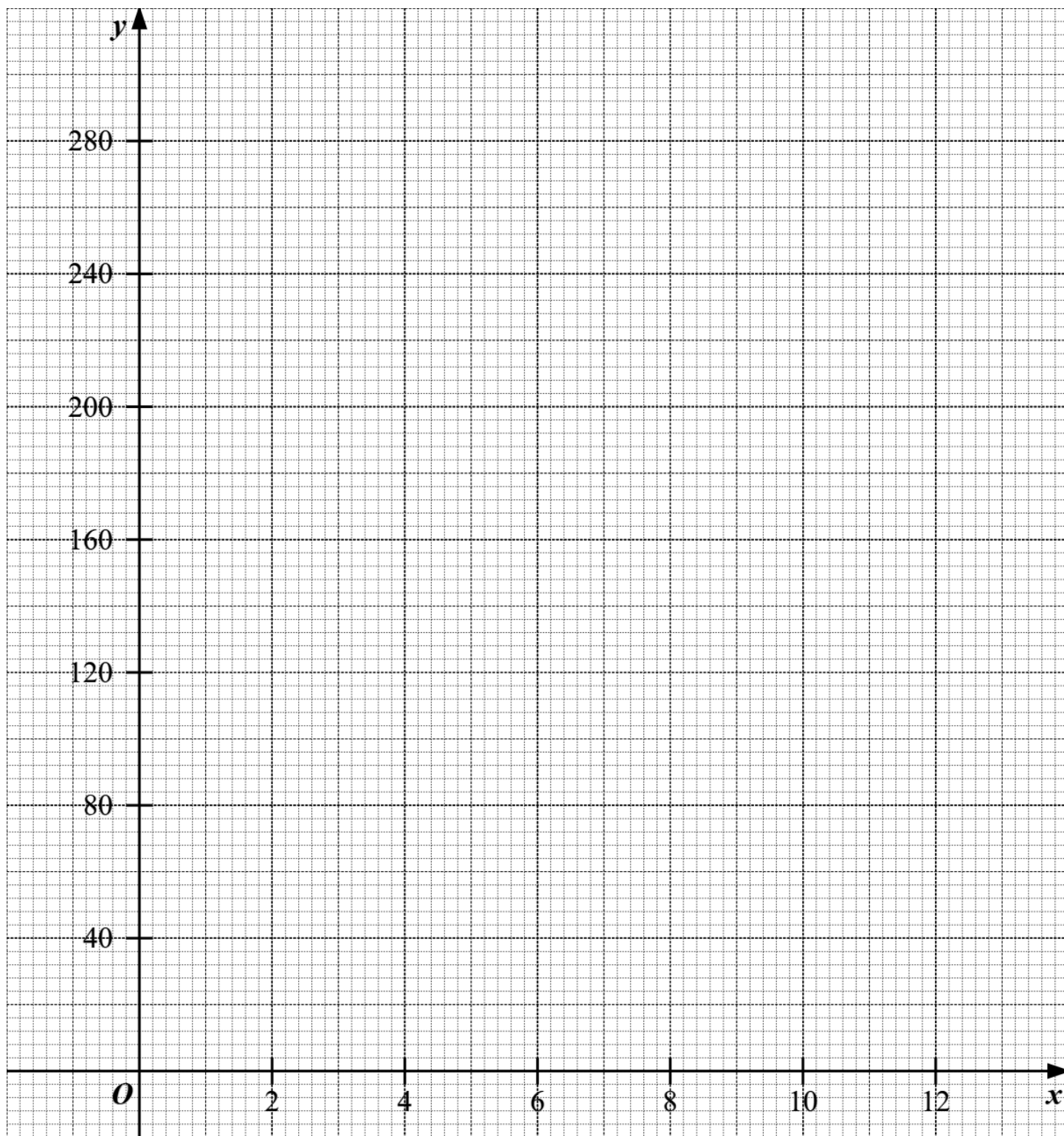
[1]

(ii) On the grid, draw the graph of $y = x\left(49 - \frac{x^2}{4}\right)$ for $0 \leq x \leq 12$.

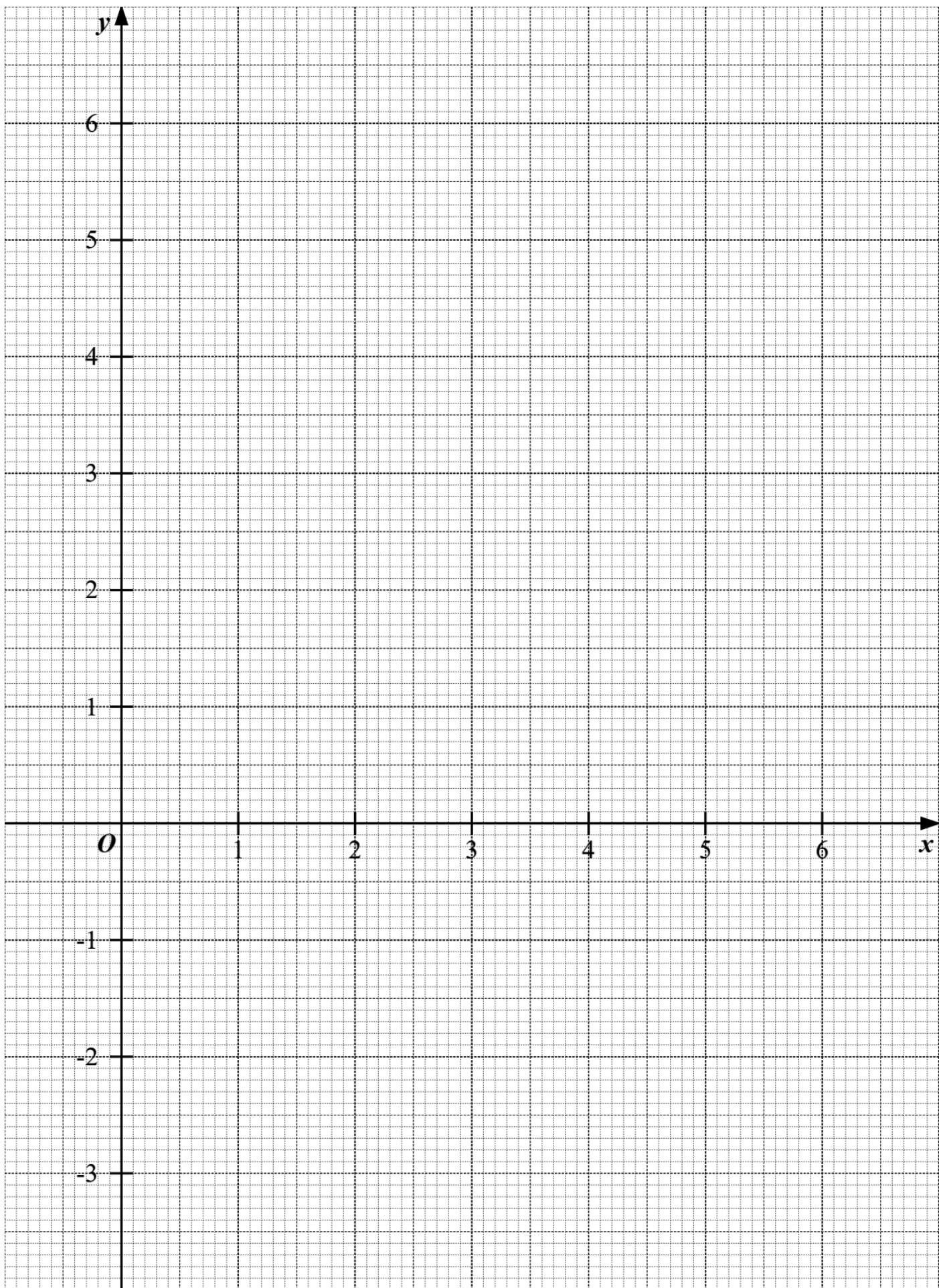
[3]

(iii) The line $y = kx + 120$, where k is a constant, is a tangent to the curve.
By drawing a suitable straight line on the graph, find the value of k .

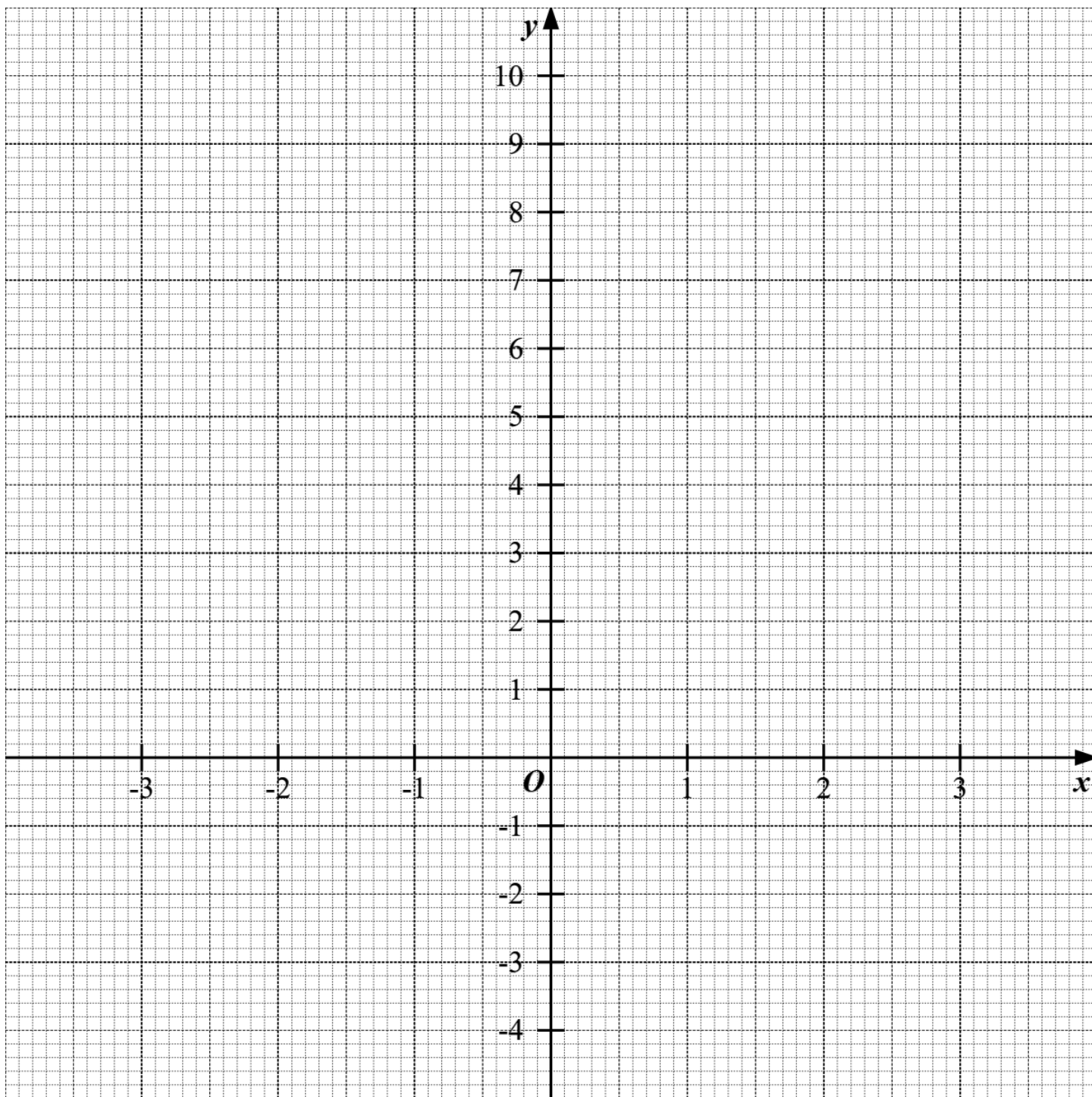
[2]



6	SST/2025/II/04																		
	(a) Complete the table of values for $y = \frac{x^2}{7} + \frac{2}{x} - 5$. Values are given to one decimal place where appropriate. <table><tr><td>x</td><td>0.2</td><td>0.5</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td></td><td>-1.0</td><td>-2.9</td><td>-3.4</td><td>-3.0</td><td>-2.2</td><td>-1.0</td><td>0.5</td></tr></table> (b) On the grid opposite, draw the graph of $y = \frac{x^2}{7} + \frac{2}{x} - 5$ for $0 < x \leq 6$. (c) Use your graph to write down an inequality in x to describe the range of values where $y < -2$. (d) (i) On the same grid, draw the graph of $2y + 4x = 9$ for $0 < x \leq 6$. (ii) Write down the x-coordinates of the points where the line intersects the curve. (iii) These values of x are the solutions of the equation $2x^3 + Ax^2 - Bx + 28 = 0$. Find the value of A and the value of B.	x	0.2	0.5	1	2	3	4	5	6	y		-1.0	-2.9	-3.4	-3.0	-2.2	-1.0	0.5
x	0.2	0.5	1	2	3	4	5	6											
y		-1.0	-2.9	-3.4	-3.0	-2.2	-1.0	0.5											



7	GMSS/2025/II/04																
(a)	Complete the table of values for $y = 0.5x^3 + 0.5x^2 - 4x + 3$.																
	<table><tr><td>x</td><td>-3</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>y</td><td></td><td>9</td><td>7</td><td>3</td><td>0</td><td>1</td><td>9</td></tr></table>	x	-3	-2	-1	0	1	2	3	y		9	7	3	0	1	9
x	-3	-2	-1	0	1	2	3										
y		9	7	3	0	1	9										
(b)	On the grid, draw the graph of $y = 0.5x^3 + 0.5x^2 - 4x + 3$ for $-3 \leq x \leq 3$.																
(c)	The equation $0.5x^3 + 0.5x^2 - 4x + 3 = k$ has one solution and k is positive.																
(i)	Draw the line $y = k$ on the same grid.																
(ii)	State the value of k .																
(d)	By drawing a tangent, find the gradient of the curve at (2, 1).																
(e)	By drawing a suitable line on the grid, solve the equation $-x^3 - x^2 + 4x + 4 = 0$.																



- (a) Complete the table of values for $y = 3x + \frac{5}{x} - 8$.

Give your answer correct to 1 decimal place.

x	0.3	0.5	1	2	3	4	5	6
y	9.6	3.5	0	0.5		5.3	8	10.8

[1]

- (b) On the grid, draw the graph of $y = 3x + \frac{5}{x} - 8$ for $0.3 \leq x \leq 6$.

[3]

- (c) Use your graph to find the solutions of the equation $3x + \frac{5}{x} = 12$ in the range $0.3 \leq x \leq 6$.

[2]

- (d) (i) On the grid in part (b), draw the line $2y = 3x - 1$ for $0 \leq x \leq 6$.

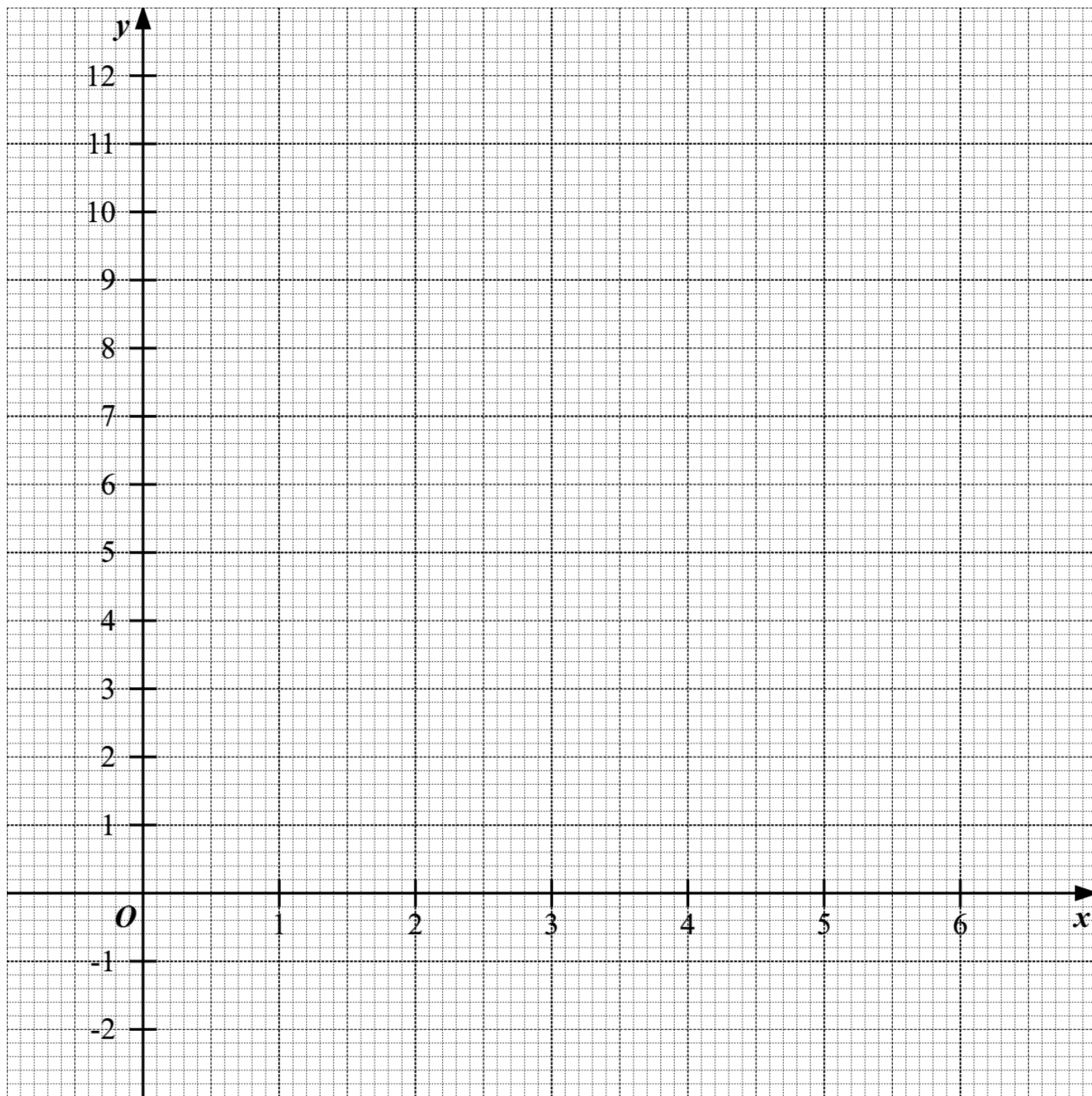
[2]

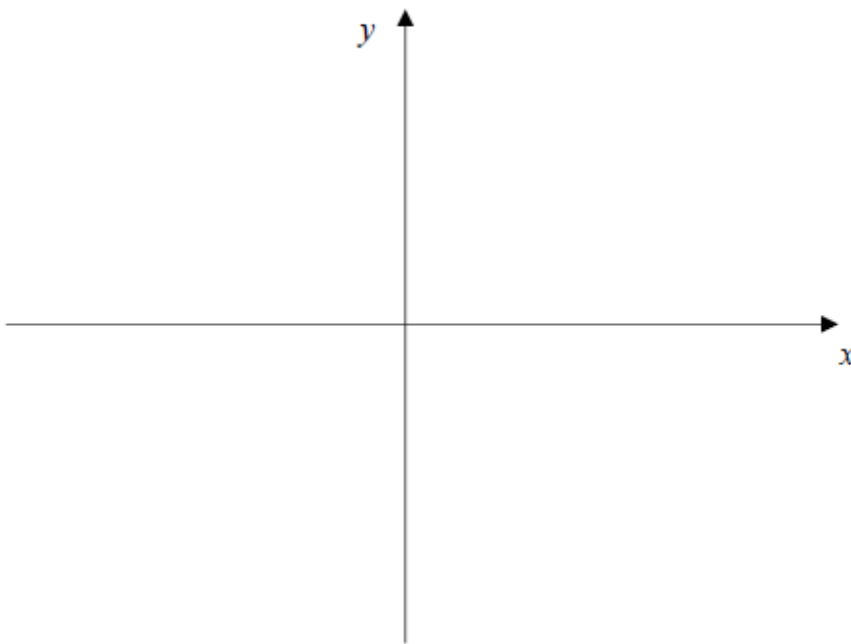
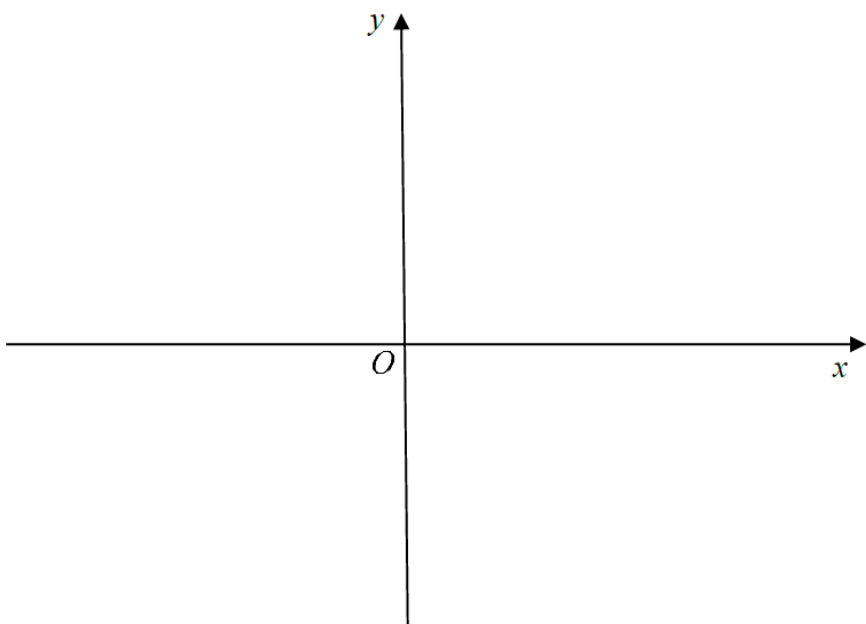
- (ii) Write down the x -coordinates of the points where this line intersects the curve.

[2]

- (iii) These values of x are the solutions of the equation $3x^2 + Ax + B = 0$. Find the value of A and the value of B .

[3]



9	ACSBR/2025/I/12
	Sketch the graph of $y = -(x+2)(5-x)$ on the axes below. Indicate clearly the points where the graph crosses the axes.  [2]
10	PHS/2025/I/15
	(a) Sketch the graph of $y = (x+4)(x-8)$ on the axes below. Indicate clearly the values where the graph crosses the x and y axes.  [2] (b) Use the graph in part (a) to explain whether the graph of $y = (x+4)(x-8) + 26$ cuts the x-axis. [1]

Solution

1	(a) $(x - 2)(-x + 6) = 0$ $x = 2 \quad \text{or} \quad x = 6$ $A = (2, 0)$ $B = (6, 0)$ (b) Line of symmetry, $x = 4$ $\text{Max } y \text{ value} = -4^2 + 8(4) - 12 = 4$ $\text{Maximum point} = (4, 4)$ (c) Since the graph <u>will not intersect the line $y = k$ when $k > 4$</u>, there are no solutions when $k > 4$. OR As $-x^2 + 8x - 12$ <u>will not go above its maximum value of 4</u>, it will not have solutions for $k > 4$.
2	(a) Using $P(0, 3)$, $k = 3$ Using $Q(-3, 192)$, $192 = 3a^{-3}$ $a^{-3} = 64$ $a = 0.25$ (b) Rick is wrong. Equation of PQ is $y = -63x + 3$. Sub $x = -1$, $y = 66$, not 8. Hence R does not lie on the line PQ. So P, Q and R are not collinear.

3

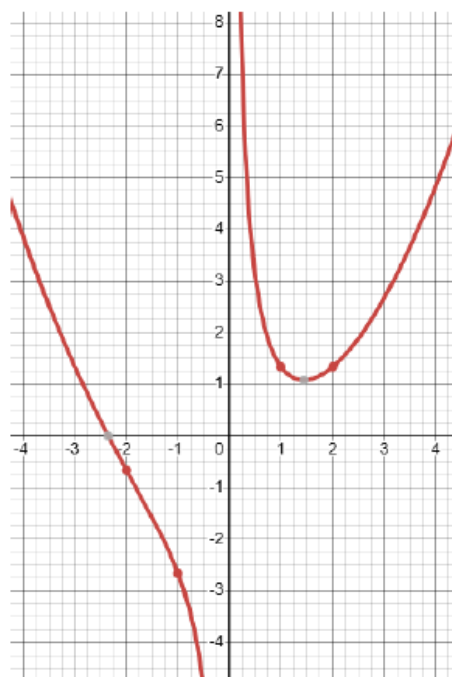
(a)

1.3

[M1] for points plotted

[M1] for RHS

[M1] for LHS



(b)

(c)

$p < 1.1$

(di)

$$2x^3 - 3x^2 - 12x + 12 = 0$$

$$\frac{2x^3 - 3x^2 - 12x + 12}{x} = 0$$

$$2x^2 - 3x - 12 + \frac{12}{x} = 0$$

$$\frac{2x^2 - 3x - 12 + \frac{12}{x}}{6} = 0$$

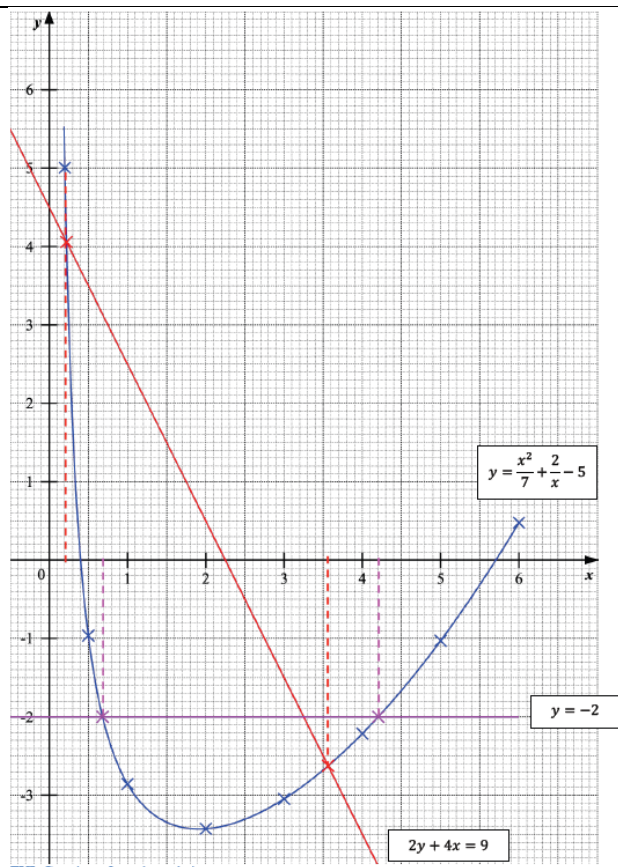
$$\frac{x^2}{3} + \frac{2}{x} - 1 = 1 + \frac{1}{2}x$$

(dii)

[M1] for drawing straight line

$$x = -2.3, 0.9, 2.9 \text{ [A1 each]}$$

4	a 2.85 b Smooth curve through 9 correct points. ci Ruled straight line through $(-2, -1)$ and $(2, 5)$. cii Accept: 0.3 / 0.35 / 0.4 / 0.45 / 0.5 or Accept: 2.8 / 2.85 / 2.9 / 2.95 / 3 d $2x^3 - 10x^2 + 35 = 0$ $-\frac{2x^3}{5} + 2x^2 - 7 = 0$ $-\frac{2x^3}{5} + 2x^2 - 3 = 4$ $\therefore y = 4$
5	(ai) figure 5 (aii) Figure 4 (aiii) Figure 3 (bi) 240 (bii) (biii) $k = \frac{240 - 160}{6 - 2}$ $= 20$
6	(a) 5.0 (1 d.p.) (b) [points plotted correctly] [smoothness of curve and label]



(c)

$$0.7 < x < 4.2 \ (\pm 0.1)$$

(di)

[at least 3 plotted points]

[graph and label]

(dii)

$$0.2, 3.55 \ (\pm 0.1)$$

(diii)

$$\frac{x^2}{7} + \frac{2}{x} - 5 = -2x + \frac{9}{2}$$

[equate both functions]

$$2x^3 + 28 - 70x = -28x^2 + 63x$$

$$2x^3 + 28x^2 - 133x + 28 = 0$$

$$A = 28, \quad B = 133$$

7

1a	6
ci	Draw the line above $y = 9$
cii	Accept any value above 9
1d	$m = 4 \pm 1$

e

$$-x^3 - x^2 + 4x + 4 = 0$$

$$\frac{-x^3 - x^2 + 4x + 4}{-2} = 0$$

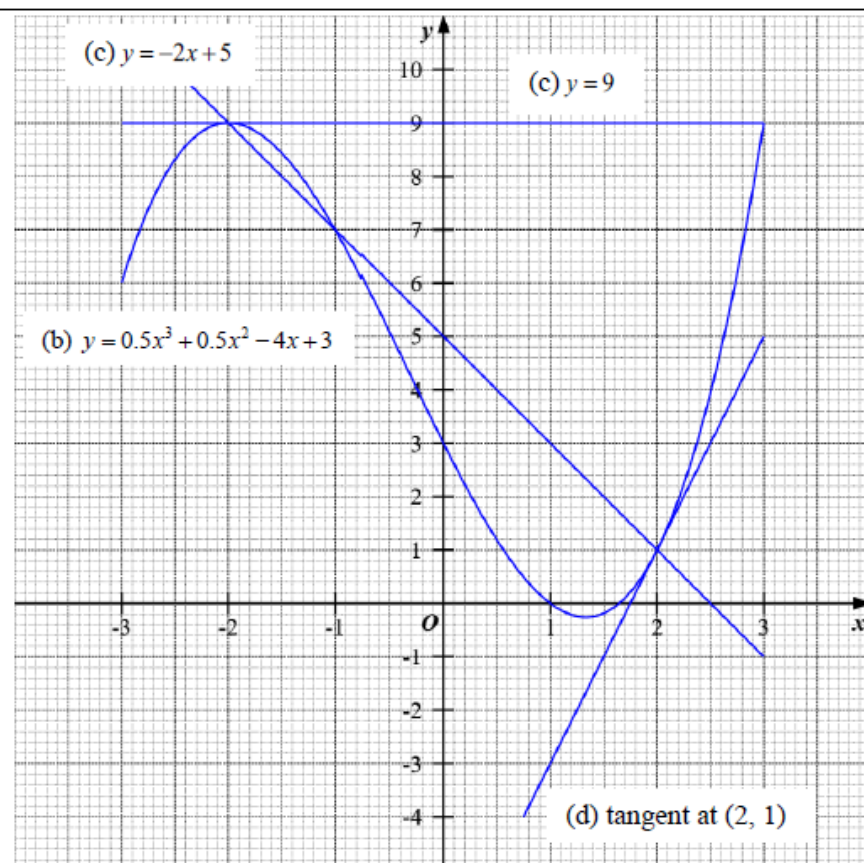
$$0.5x^3 + 0.5x^2 - 2x - 2 = 0$$

$$0.5x^3 + 0.5x - 4x + 3 = -2x + 5$$

Draw the line $y = -2x + 5$

From the graph, $x = -2, -1, 2$

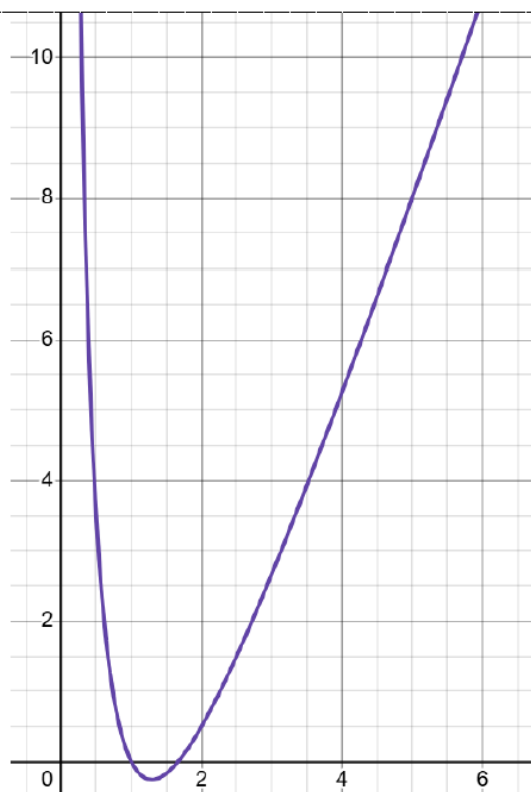
b



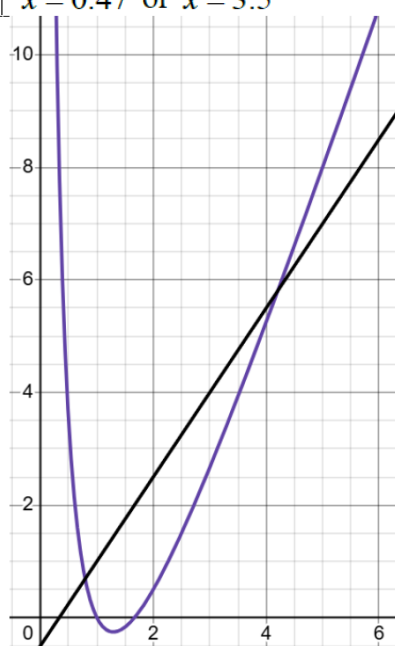
8

(a) 2.7

(b)

(c) $x = 0.47$ or $x = 3.5$

(di)

(dii) $x = 0.8$ or $x = 4.2$

(diii) $6x + \frac{10}{x} - 16 = 3x - 1$

$$6x^2 + 10 - 16x = 3x^2 - x$$

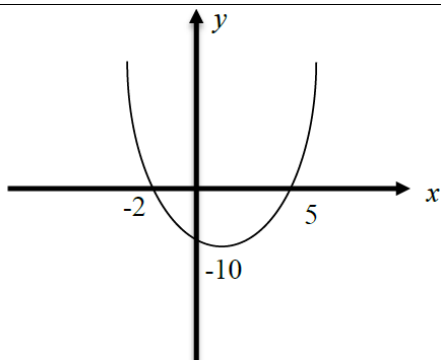
By comparing coefficients,

$$3x^2 - 15x + 10 = 0$$

$$A = -15$$

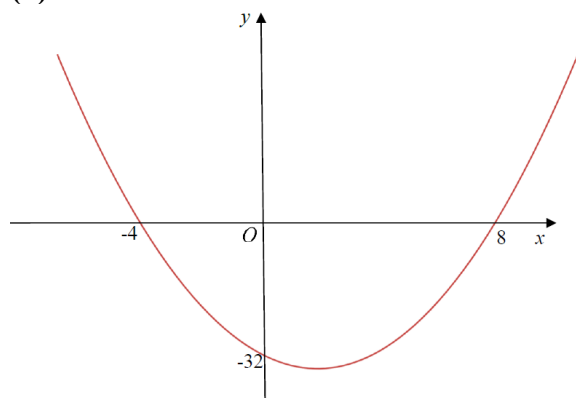
$$B = 10$$

9



10

(a)



(b)

y -intercept of $y = (x+4)(x-8)$ is -32 , therefore y -intercept of $y = (x+4)(x-8) + 26$ is -6 which is below the x -axis, therefore y **cuts** the x -axis at two points.

Or

Minimum value of $y = (x+4)(x-8)$ is -36 , therefore **minimum** value of

$y = (x+4)(x-8) + 26$ is -10 which is below the x -axis, therefore y **cuts** the x -axis at two points because its vertex lies below the x -axis.